

AS ECONOMICS 7135/1

Paper 1 The Operation of Markets and Market Failure

Practice Paper — Advanced

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Insert (Extracts A–F included within this paper)
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Answer **all** questions in **Section A**.
- Answer **either** Context 1 **or** Context 2 in **Section B**.
- You must answer the questions in the spaces provided.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The maximum mark for this paper is 70.
- The marks for questions are shown in brackets.
- No deductions will be made for wrong answers.

Section A

Answer **all** questions in this section. Only **one** answer per question is allowed.

01 The fundamental purpose of the price mechanism in a free market is to

- A** ensure an equitable distribution of income and wealth.
- B** ration scarce resources between competing uses.
- C** guarantee minimum standards of consumer protection.
- D** set prices according to government targets.

[1 mark]

02 Which one of the following is most clearly an assumption of the rational consumer model?

- A** Consumers always have imperfect information.
- B** Consumers seek to maximise their utility given their budget.
- C** Consumers are influenced primarily by herd behaviour.
- D** Consumers' tastes change unpredictably from day to day.

[1 mark]

03 Figure 1 shows the original demand curve *D* for a normal good. The demand curve subsequently shifts to *D1*.

[1 marks]

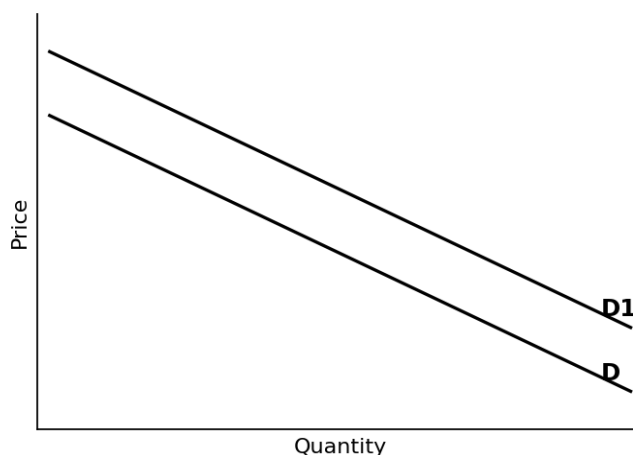


Figure 1

Which one of the following is most likely to have caused this shift?

- A** A fall in consumer incomes.
- B** A rise in the price of a complementary good.
- C** A successful advertising campaign for the good.
- D** An increase in the cost of producing the good.

[1 mark]

04 Which one of the following is most likely to be the long-run consequence of a market with significant barriers to entry?

- A** Many firms entering and earning normal profit only.
- B** Incumbent firms sustaining supernormal profits.
- C** Allocative efficiency at $MC = P$.
- D** Prices being driven below average total cost.

[1 mark]

05 The price elasticity of demand for a public transport service is -0.3 . If fares rise by 10%, the most likely impact on total revenue is

- A** an increase of approximately 7%.
- B** an increase of approximately 13%.
- C** a fall of approximately 3%.
- D** no change.

[1 mark]

06 Which one of the following best describes a positive consumption externality?

- A** Smoking imposing health costs on non-smokers in restaurants.
- B** A neighbour playing music too loud at night.
- C** Vaccination reducing the risk of disease for non-vaccinated individuals.
- D** A firm dumping chemicals into a river.

[1 mark]

07 An indirect ad valorem tax of 20% is placed on a good. Compared with a specific (per unit) tax of equivalent value, the ad valorem tax

- A** shifts the supply curve to the left by the same amount at every price.
- B** raises the equilibrium price by a higher proportion at low price levels.
- C** pivots the supply curve, raising it by a larger amount at higher prices.
- D** always raises more revenue regardless of demand conditions.

[1 mark]

08 A firm's average total cost is £40 and its marginal cost is £30. Producing one more unit will

- A** increase average total cost.
- B** decrease average total cost.
- C** leave average total cost unchanged.
- D** increase total fixed cost.

[1 mark]

09 Table 1 shows four features of a market.

[1 marks]

Feature	Yes / No
Many small firms each producing a small fraction of total output	Yes
Homogeneous (identical) products	Yes
Free entry and exit	Yes
Perfect information available to buyers and sellers	Yes

Table 1

Which market structure is most consistent with the features shown?

- A** Monopoly.
- B** Oligopoly.
- C** Monopolistic competition.
- D** Perfect competition.

[1 mark]

10 Which one of the following is most likely to cause a parallel rightward shift of the supply curve for wheat?

- A** A rise in the price of wheat.

- B A government subsidy paid per tonne of wheat produced.
- C A fall in the price of barley, an alternative crop.
- D Adverse weather conditions during the growing season.

[1 mark]

11 The marginal external cost of producing steel exceeds the marginal external benefit. The free-market output of steel is therefore

- A below the socially optimal level.
- B above the socially optimal level.
- C at the socially optimal level.
- D impossible to determine.

[1 mark]

12 Information failure in the market for second-hand cars (the 'lemons' problem) is best described as a situation where

- A buyers know more than sellers about the quality of cars.
- B sellers know more than buyers about the quality of cars.
- C neither buyers nor sellers can observe the price of cars.
- D the government deliberately restricts information.

[1 mark]

13 Figure 2 shows the demand curve for a good. The firm currently sells the good at a price of £8.

[1 marks]

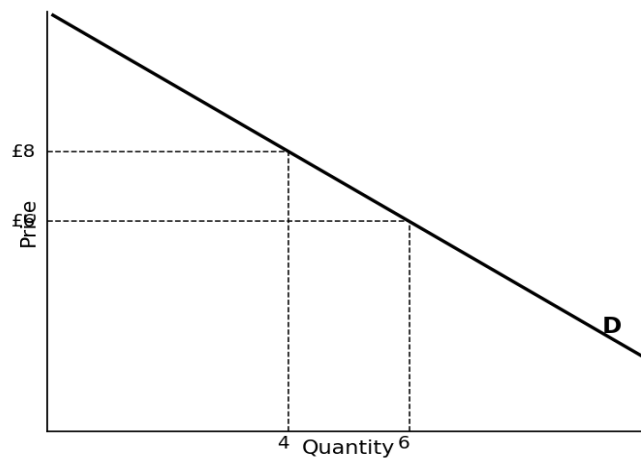


Figure 2

If the firm cuts the price to £6, what happens to total revenue?

- A Total revenue rises from £32 to £36.
- B Total revenue rises from £32 to £48.
- C Total revenue falls from £32 to £24.
- D Total revenue rises from £24 to £48.

[1 mark]

14 A minimum wage set above the equilibrium wage in the labour market is most likely to cause

- A a shortage of labour as quantity demanded exceeds quantity supplied.
- B an excess supply of labour at the minimum wage.
- C an increase in employment of low-skilled workers.
- D the market to clear at the new wage.

[1 mark]

15 A firm experiences diseconomies of scale. This is most likely to be shown on a long-run average cost curve diagram as

- A** the downward-sloping section to the left of the minimum.
- B** the horizontal section at the minimum efficient scale.
- C** the upward-sloping section to the right of the minimum.
- D** a horizontal line at the level of normal profit.

[1 mark]

16 Cross elasticity of demand between Good X and Good Y is -1.2 . This means that

- A** X and Y are weak complements.
- B** X and Y are strong complements.
- C** X and Y are weak substitutes.
- D** X and Y are strong substitutes.

[1 mark]

17 Which one of the following statements about productivity is most likely to be true?

- A** Higher labour productivity necessarily means fewer workers are employed.
- B** Higher labour productivity tends to reduce the unit labour cost of production.
- C** Higher labour productivity always reduces total output.
- D** Higher labour productivity is always achieved by reducing capital investment.

[1 mark]

18 An income elasticity of demand of $+2.4$ for foreign holidays means that foreign holidays are

- A** an inferior good.
- B** a necessity.
- C** a normal luxury good.
- D** a Giffen good.

[1 mark]

19 Tradable pollution permits attempt to correct market failure by

- A** increasing the supply of polluting goods.
- B** creating a market in the right to pollute and capping total pollution.
- C** subsidising firms that pollute the most.
- D** abolishing externalities altogether.

[1 mark]

20 A merit good is best defined as a good that

- A** is consumed only by high-income individuals.
- B** would be under-provided and under-consumed in a free market because consumers undervalue the long-term private and social benefits.
- C** is non-rival and non-excludable in consumption.
- D** has zero marginal cost of production.

[1 mark]

END OF SECTION A

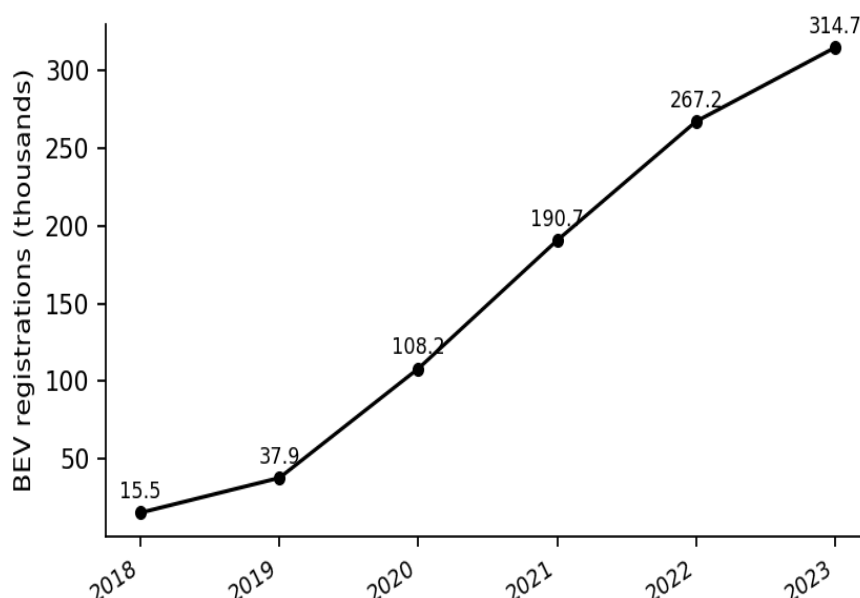
Section B

Answer **either** Context 1 **or** Context 2. There is a total of 50 marks for either context.

EITHER

Context 1 — The market for electric vehicles (EVs) in the UK *Total: 50 marks*

Extract A: New battery electric vehicle (BEV) registrations in the UK, 2018–2023 (thousands)



Source: adapted from SMMT data

Extract B: A market in transition

- 1 Sales of new battery electric vehicles in the UK have grown rapidly. Falling battery
- 2 costs, government grants for buyers, exemptions from road tax and the expansion
- 3 of the charging network have encouraged consumers to switch from petrol and
- 4 diesel cars. EV manufacturers benefit from significant economies of scale.
- 5 Larger production volumes have allowed firms to lower the average cost of each
- 6 car produced, helping prices to fall for some models.
- 7 EVs generate fewer emissions than internal combustion engine vehicles during use.
- 8 The reduction in air pollution is a **positive externality** of consumption, creating
- 9 social benefits that include lower healthcare costs and improved quality of life.
- 10 However, the production of batteries requires lithium, cobalt and other minerals,
- 11 the mining of which can damage the environment in producing countries. The end
- 12 of life disposal of batteries also raises environmental concerns.

Extract C: Government policy and the road ahead

- 1 The UK government has set a target for all new cars sold to be zero emission by
- 2 2035. From 2024, manufacturers face a Zero Emission Vehicle (ZEV) mandate
- 3 requiring an increasing percentage of their new car sales to be electric. Firms
- 4 that miss the target face fines of £15 000 per non-compliant car. Critics warn
- 5 that the policy could make car ownership less affordable for those on low incomes,
- 6 as EVs cost more than equivalent petrol cars to buy. There is also concern that
- 7 rural households have limited access to public chargers and may be at risk of being
- 8 'left behind'. The government has begun to phase out the plug-in car grant, while
- 9 continuing to subsidise public charge points and to fund research into battery
- 10 technology. Other suggested measures include scrappage schemes for older polluting
- 11 vehicles, congestion charges on petrol/diesel cars in city centres, and tougher

12 regulation of advertising for high-emission cars.

13 Should the government do more to accelerate the shift to EVs, or are there more

14 urgent priorities?

21 Define '**positive externality**' (**Extract B**, line 8).

[3 marks]

22 **Extract A** shows BEV registrations from 2018 to 2023. Calculate the percentage change in BEV registrations between 2020 and 2023. Give your answer to **one** decimal place.

[4 marks]

23 Use **Extract A** to identify **two** significant features of UK BEV registrations over the period shown.

[4 marks]

24 An economy uses limited resources to produce two products, 'electric vehicles' and 'other goods and services'. Draw a production possibility diagram showing the potential output of these two products. Show on this diagram the effect of an increase in productivity in EV manufacturing only.

[4 marks]

25 **Extract B** (lines 4–6) refers to economies of scale in EV production. Explain how economies of scale could affect the price of electric vehicles in the UK.

[10 marks]

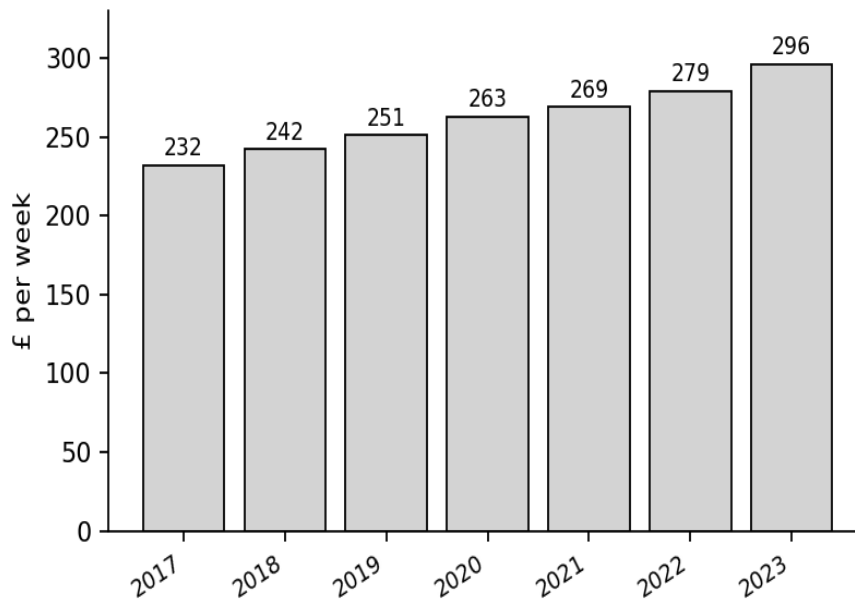
26 **Extract C** (lines 13–14) asks whether the government should do more to accelerate the shift to EVs. Use the extracts and your knowledge of economics to discuss whether the UK government should provide more support for the EV market.

[25 marks]

OR

Context 2 — Childcare in the UK Total: 50 marks

Extract D: Average weekly cost of full-time nursery childcare for a child under 2 in England (£ per week)



Source: adapted from Coram Family and Childcare survey

Extract E: The childcare market in the UK

1 The UK has one of the most expensive childcare markets in the world. Parents of
2 under-2s pay an average of £296 per week for full-time nursery care. The number
3 of registered childcare providers in England fell from 81 300 in 2015 to 68 100 in
4 2023, a fall of 16%. Many small private nurseries report that they are operating
5 close to or below cost, citing rising staff wages, food prices and energy bills as
6 **variable costs** that are difficult to control.
7 Government support has expanded over time. Eligible families can claim 15 or 30
8 hours of 'free' childcare per week during term time. From 2024, this has been
9 extended to working parents of children aged 9 months and above. Tax-Free
10 Childcare offers parents up to £2 000 a year per child to subsidise costs. However,
11 providers complain that the hourly rate paid by the government for 'free' hours does
12 not cover the cost of provision, forcing nurseries to charge higher fees for additional
13 paid hours.

Extract F: Wider economic effects

1 Access to affordable childcare matters for the wider economy. High childcare
2 costs can deter parents — usually mothers — from returning to paid work or
3 from working full-time. This contributes to the gender pay gap and a reduction in
4 the size of the labour force. Productivity may also suffer as workers' skills become
5 outdated during long career breaks.
6 Early years education is sometimes described as a merit good, with substantial
7 long-term benefits including better academic outcomes, improved life chances
8 and higher future earnings for children. These benefits accrue not only to the
9 child and family but to wider society. Some commentators describe early years
10 education as one of the best long-term investments a country can make.
11 Critics of additional state intervention argue that the existing schemes are already
12 costly, may not raise the quality of provision and that other priorities — such as
13 the NHS or schools — should come first.

27 Define '**variable costs**' (**Extract E**, line 6).

[3 marks]

28 Extract E (lines 3–4) states that the number of registered childcare providers in England fell from 81 300 in 2015 to 68 100 in 2023. Calculate the percentage change in the number of providers over this period. Give your answer to **one** decimal place.

[4 marks]

29 Use **Extract D** to identify **two** significant features of the average weekly cost of full-time nursery childcare for a child under 2 over the period shown.

[4 marks]

30 Workers in the labour force allocate their time between two activities, 'paid work' and 'unpaid childcare'. Draw a production possibility diagram showing the trade-off between these two activities. Show on this diagram the effect of an increase in the availability of subsidised paid childcare.

[4 marks]

31 Extract E (lines 1–6) refers to rising costs and falling numbers of childcare providers. Explain factors that could cause a rise in the price of childcare.

[10 marks]

32 Extract F (lines 11–13) refers to critics' arguments that other priorities should come first. Use the extracts and your knowledge of economics to discuss whether the UK government should increase its subsidies for childcare.

[25 marks]

END OF QUESTIONS